EKS Session

Summary 15-4-2023

- When you use eksctl to create an Amazon EKS cluster, it uses AWS **CloudFormation** in the background to provision the necessary resources for the cluster. eksctl abstracts away many of the underlying CloudFormation details, making it easier and faster to create an EKS cluster.
- eksctl uses **CloudFormation stacks** to create and manage the underlying resources needed for the EKS cluster, such as VPCs, subnets, security groups, IAM roles, and Amazon EKS service roles.
- It also creates the necessary Kubernetes resources, such as the control plane and worker node groups.
- The use of CloudFormation behind the scenes in eksctl simplifies the process of creating and managing the EKS cluster.
- There are multiple ways to launch a Kubernetes cluster on AWS such as using **YAML code, command-line tools, and CloudFormation code**.
- **Command-line** tools like eksctl provide an easy-to-use interface for launching Kubernetes clusters on AWS.
- Using **YAML code** to launch a Kubernetes cluster is the most flexible option, as it allows you to define your Kubernetes resources in a granular way. You have full control over the configuration of your cluster and can define your resources exactly as you need them.
- You can directly use **CloudFormation code** to launch a Kubernetes cluster and manage the cluster.

- However, the command-line may be less flexible than using YAML code, as they may not provide access to all of the configuration options available in Kubernetes, and like YAML code, using CloudFormation may require a deep understanding of AWS and its services.

- By default eksctl create cluster will create a dedicated VPC for the cluster.

- The VPC is a logical isolated network in Amazon Web Services (AWS). Each subnet within the VPC has a CIDR block associated with it, which defines the range of IP addresses that can be used within the subnet.

- Worker nodes and pods in an EKS cluster are assigned IP addresses from the subnets associated with the Kubernetes service.

- To launch an Amazon Elastic Kubernetes Service (EKS) cluster using YAML code, you can run the following command:

    - Download the YAML code:

```
C:\Users\Vimal Daga\Documents\eks_advance_code>curl https://raw.githubusercontent.com/weavewo
rks/eksctl/main/examples/01-simple-cluster.yaml -o 01-simple-cluster.yaml
```

    - Create the cluster:

```
C:\Users\Vimal Daga\Documents\eks_advance_code>eksctl create cluster -f 01-simple-cluster.yam
l
```

- On the eksctl website, you can find a section of the "Examples". If you click on "Examples", you will land on the GitHub page. There you can find a repository containing a variety of example YAML configurations for different scenarios. These examples include YAML code for creating a simple cluster, a cluster with spot instances, a cluster with multiple node groups, creating an EKS cluster with Fargate, a cluster with a custom VPC, a cluster with multiple node groups and managed node groups, and much more.

- Using the YAML file you can customize many things such as:

1) Enable specific types of cluster control plane logs

2) Custom VPC

3) Auto-allocated IPv6 CIDRs for all subnets

4) Disable public access to the endpoint and only allow private access

5) Create multiple node-groups

6) Launch Kubernetes cluster on existing VPC

7) More customized node groups: eg. set label to node, minSize or maxSize for node, etc.

8) Access to CSI drivers

- To set the label of the node:

```
C:\Users\Vimal Daga>kubectl  label node ip-192-168-34-140.ap-south-1.compute.internal   env=dev
node/ip-192-168-34-140.ap-south-1.compute.internal labeled
```

- To see the labels of the node:

```
C:\Users\Vimal Daga>kubectl get nodes --show-labels
NAME                                           STATUS   ROLES    AGE   VERSION               LABELS
ip-192-168-34-140.ap-south-1.compute.internal   Ready    <none>   52m   v1.24.11-eks-a59e1f0   alpha.eksctl.io/cluster-
name=cluster2,alpha.eksctl.io/nodegroup-name=ng-142eb352,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-type
=m5.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON_DEMAND,eks.amazonaws.com/nodegroup-image=ami-0a
6844ad645e3b8c7,eks.amazonaws.com/nodegroup=ng-142eb352,eks.amazonaws.com/sourceLaunchTemplateId=lt-080eedfd1dcf0841d,e
ks.amazonaws.com/sourceLaunchTemplateVersion=1,env=dev,failure-domain.beta.kubernetes.io/region=ap-south-1,failure-doma
in.beta.kubernetes.io/zone=ap-south-1a,k8s.io/cloud-provider-aws=ddea69abfa1ba4e22536279ae15c35b7,kubernetes.io/arch=am
d64,kubernetes.io/hostname=ip-192-168-34-140.ap-south-1.compute.internal,kubernetes.io/os=linux,node.kubernetes.io/inst
ance-type=m5.large,topology.kubernetes.io/region=ap-south-1,topology.kubernetes.io/zone=ap-south-1a
ip-192-168-92-51.ap-south-1.compute.internal    Ready    <none>   52m   v1.24.11-eks-a59e1f0   alpha.eksctl.io/cluster-
name=cluster2,alpha.eksctl.io/nodegroup-name=ng-142eb352,beta.kubernetes.io/arch=amd64,beta.kubernetes.io/instance-type
=m5.large,beta.kubernetes.io/os=linux,eks.amazonaws.com/capacityType=ON_DEMAND,eks.amazonaws.com/nodegroup-image=ami-0a
6844ad645e3b8c7,eks.amazonaws.com/nodegroup=ng-142eb352,eks.amazonaws.com/sourceLaunchTemplateId=lt-080eedfd1dcf0841d,e
ks.amazonaws.com/sourceLaunchTemplateVersion=1,env=prod,failure-domain.beta.kubernetes.io/region=ap-south-1,failure-dom
ain.beta.kubernetes.io/zone=ap-south-1c,k8s.io/cloud-provider-aws=ddea69abfa1ba4e22536279ae15c35b7,kubernetes.io/arch=a
md64,kubernetes.io/hostname=ip-192-168-92-51.ap-south-1.compute.internal,kubernetes.io/os=linux,node.kubernetes.io/inst
ance-type=m5.large,topology.kubernetes.io/region=ap-south-1,topology.kubernetes.io/zone=ap-south-1c
```

- Node labels enable Kubernetes to schedule pods on specific nodes.

- To enable all types of logs:

```
C:\Users\Vimal Daga\Documents\eks_advance_code>eksctl utils  update-cluster-logging   --cluster cluster2  --enab
le-types all   --approve
```